Control Column – General Data

1. General

The dual control column consists of an upper and lower section. The upper section is bolted to the lower section and consists of a fork assembly, two W-type handwheels, control chains and sprockets. The control chains driven by sprockets splined to the handwheel shafts, are routed over idler sprockets in the vee of the fork through the lower upright section to a bottom sprocket at the aileron quadrant. A torque tube is welded to the lower section carrying an elevator control arm, control column bearing assemblies, and the housing for the aileron quadrant, sprocket and stop lever. A centering device, in the form of a spring strut, is attached to a lever on the sprocket shaft of the co-pilot's handwheel, to overcome circuit friction. The left arm of the upper section includes the nose wheel steering lever and cable assembly (refer to 32-50-00, Nose Wheel Steering System – General Data). A luminous tape nose wheel position indicator (Mod 6/1396) fixed to the lever and cover provides a neutral position reference. A nose wheel position indicator light (Mod 6/1478) mounted on the cover provides illumination of the indicator (refer to Figure 1).

2. Removal/Installation

Refer to Figure 1 for details.

A. Remove Control Column Assembly

1) Remove hood assembly from base of control column and flight compartment floor.

2) Gain access to under flight compartment floor by removing forward fuselage access panels.

3) Disconnect pilot's and co-pilot's microphone switch wiring at terminal block.

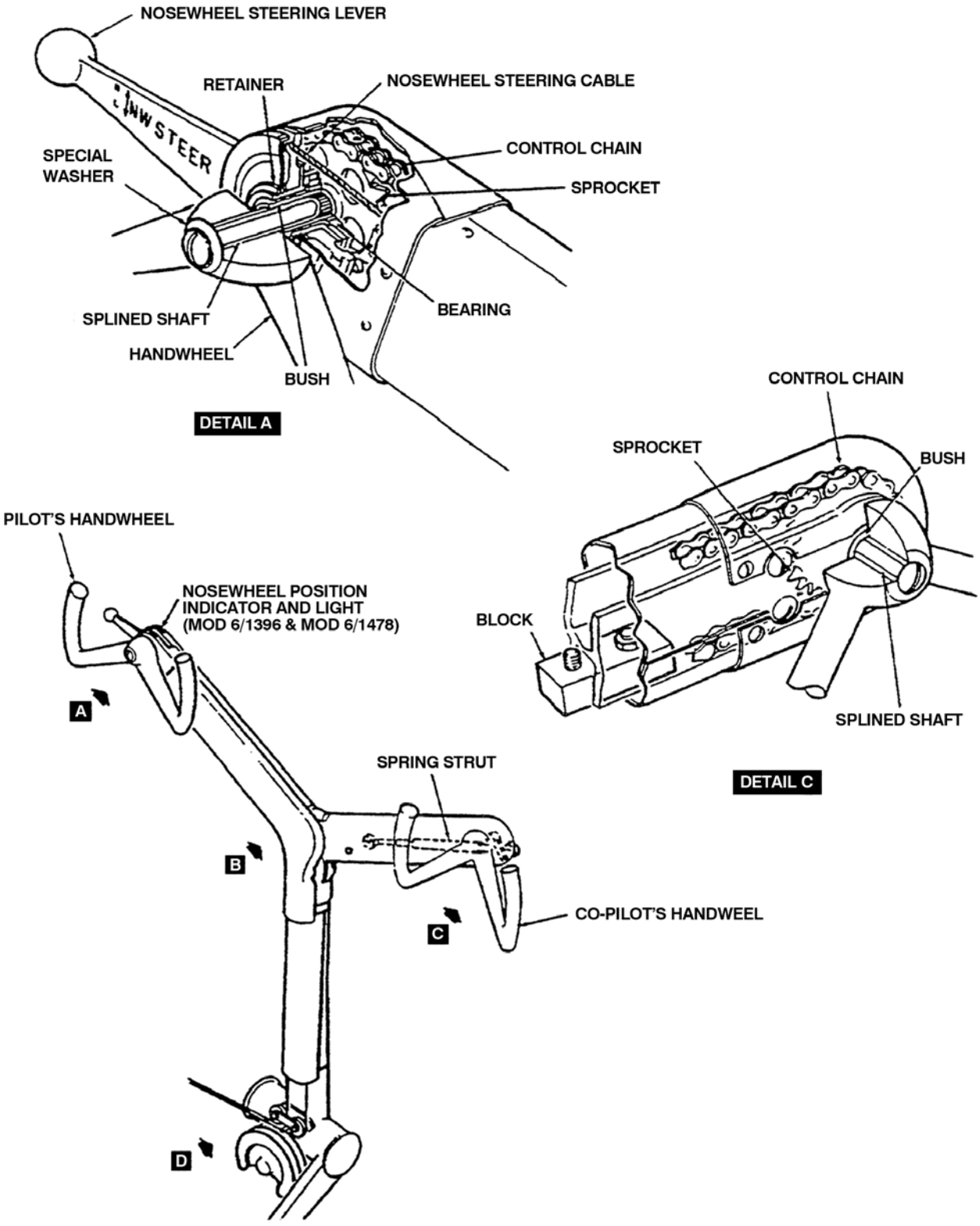
4) Disconnect nose wheel steering cables at turnbuckles and release cables from pulleys at airframe structure.

5) Remove cotter pin and pin guard from lower end of control column at aileron quadrant.

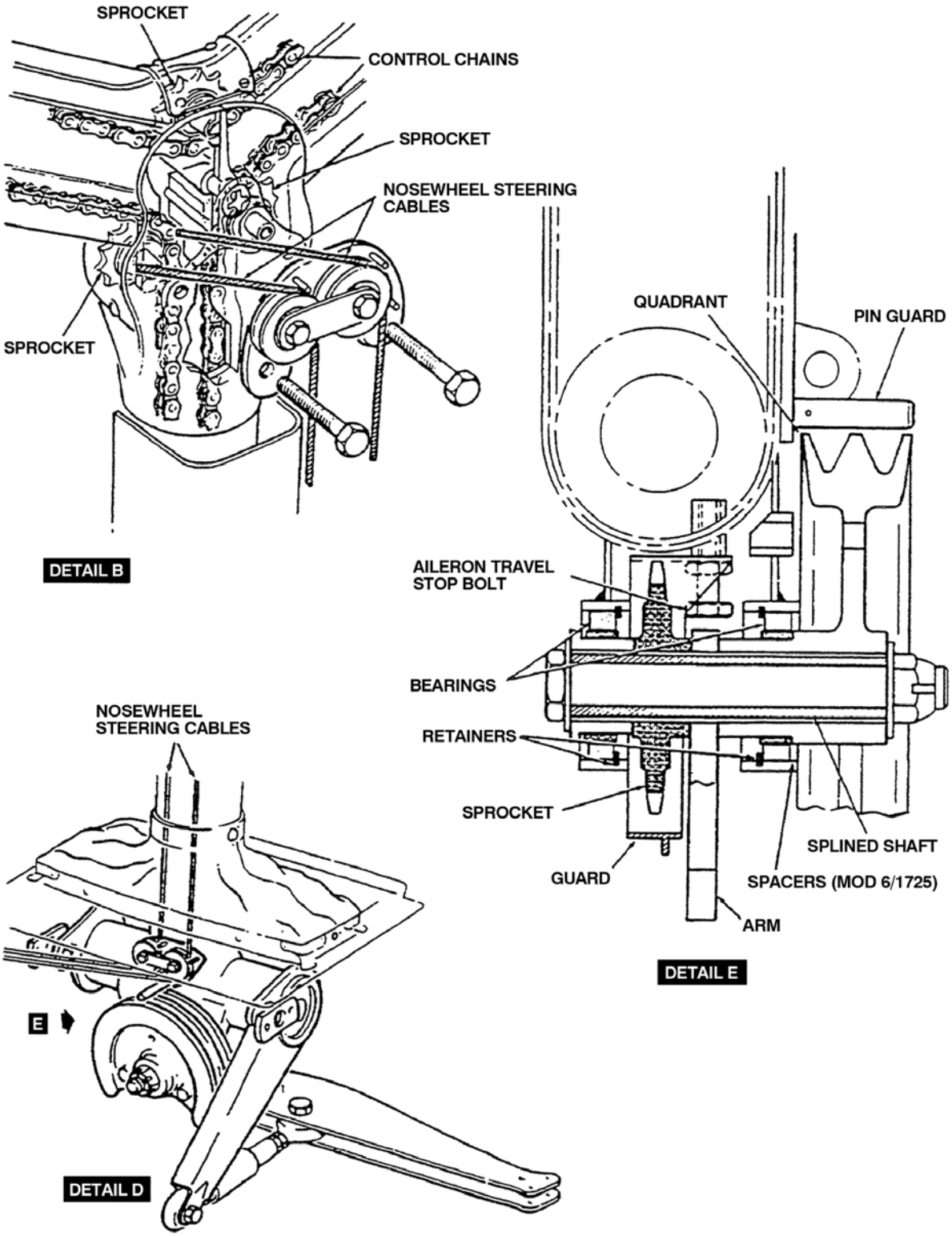
6) Slacken aileron control cables at turnbuckles and disconnect from aileron quadrant.

7) Disconnect connecting rod from control column elevator control arm.

8) Support control column and remove two bolts securing column to structure mounted housing assemblies, Remove control column from aircraft.



Control Column
Figure 1 (Sheet 1 of 2)



Control Column
Figure 1 (Sheet 2 of 2)

B. Install Control Column Assembly

Note Coat bolt shanks with corrosion preventive compound MIL–C–16173 Grade 2 or equivalent, before installation.

1) Position control column in aircraft and install bolts, washers and nuts to secure column to housing assemblies. Tighten nuts and secure with cotter pins.

2) Install bolt, washer and nut to secure connecting rod to elevator control arm. Tighten nut and secure with cotter pin.

3) Connect aileron control cables to aileron quadrant at base of control column; install pin guard and cotter pin.

4) Route nose wheel steering cables through pulleys on left side of aircraft structure, outboard of control column. If disturbed secure pulleys to bracket.

5) Connect nose wheel steering cable turnbuckles.

6) Connect pilot's and co-pilot's microphone switch cables at terminal block and test audio system (refer to 23-50-00, Audio Integration – General Data).

7) Check aileron control handwheel alignment (refer to 27-10-00, Aileron Control System – General Data).

8) Rig aileron control system (refer to 27-10-00, Aileron Control System – General Data).

9) Rig and test nose wheel steering system (refer to 32-50-00, Nose Wheel Steering System – General Data).

10) Rig elevator control system (refer to 27-30-00, Elevator Control System – General Data).

11) Secure underfloor access panels removed in Para 2.A.2).

12) Install hood assembly at base of control column.

13) Ensure control column and handwheel freedom of movement throughout fore and aft control column travel.